

AEGIS ENGINEERING HANDBOOK

Artificial Financial Intelligence Platform

Complete Engineering Documentation — Documents 01 through 10

Corrected Edition

Correction applied: Document 10, Section 17 (Architect Commitments) — Chief Systems Architect updated from ChatGPT to Claude.

Table of Contents

Doc	Title	ID
Document 01	PROJECT VISION	AEGIS-DOC-001
Document 02	GUIDING PRINCIPLES	AEGIS-DOC-002
Document 03	SYSTEM ARCHITECTURE	AEGIS-DOC-003
Document 04	COGNITIVE ARCHITECTURE	AEGIS-DOC-004
Document 05	AI COUNCIL ARCHITECTURE	AEGIS-DOC-005
Document 06	DATA FLOW & DECISION PIPELINE	AEGIS-DOC-006
Document 07	DATABASE ARCHITECTURE	AEGIS-DOC-007
Document 08	TECHNOLOGY STACK & INFRASTRUCTURE	AEGIS-DOC-008
Document 09	DEVELOPMENT ROADMAP	AEGIS-DOC-009
Document 10	MASTER PROJECT CONSTITUTION	AEGIS-DOC-010

PROJECT VISION

Document ID: AEGIS-DOC-001

Version: 1.0

Status: Approved

Classification: Core Engineering Documentation

Project: AEGIS – Artificial Financial Intelligence Platform

Executive Summary

Project AEGIS is an institutional-grade Artificial Financial Intelligence Platform engineered to observe financial markets, understand market behavior, coordinate specialized artificial intelligence systems, perform evidence-based reasoning, execute disciplined risk management, and continuously improve through scientific research.

Unlike traditional trading bots, AEGIS is designed as a complete intelligence ecosystem rather than a collection of trading algorithms. Trading represents only one capability within the platform; research, learning, explainability, and continuous adaptation form its foundation.

The platform is intended to evolve over many years into a professional financial research operating system capable of supporting multiple asset classes while maintaining engineering discipline, transparency, and scalability.

Vision Statement

To build the world's most intelligent, explainable, adaptive, and continuously improving Artificial Financial Intelligence Platform capable of assisting financial decision-making through scientific research, modular artificial intelligence, and institutional-grade risk management.

Mission Statement

Project AEGIS exists to:

- Observe financial markets continuously.
- Understand market context rather than isolated price movements.
- Coordinate multiple specialized AI experts.
- Produce explainable recommendations supported by evidence.
- Protect capital through rigorous risk management.
- Learn continuously from every market interaction.
- Preserve research permanently for future improvement.
- Create an evolving financial intelligence ecosystem rather than a static trading application.

The Problem Statement

Current retail trading systems suffer from several critical limitations:

- Static trading strategies that degrade as markets evolve.
- Heavy dependence on technical indicators alone.
- Poor understanding of macroeconomic events.
- Lack of explainable decision-making.
- Weak risk governance.
- No institutional memory.
- Limited ability to learn from previous outcomes.

These limitations reduce long-term reliability and adaptability. AEGIS is specifically designed to overcome these weaknesses.

What AEGIS Is

AEGIS is:

- An Artificial Financial Intelligence Platform.
- A Quantitative Research Environment.
- A Decision Intelligence System.
- A Continuous Learning Platform.
- A Financial Knowledge Engine.
- An Explainable AI Ecosystem.

What AEGIS Is Not

AEGIS is not:

- A simple Forex bot.
- An Expert Advisor (EA).
- A signal-selling platform.
- A collection of technical indicators.
- A price prediction machine.
- A "set-and-forget" trading robot.

The purpose of AEGIS is to improve decision quality—not blindly automate trading.

Core Philosophy

The philosophy of AEGIS is based on one principle: Consistently making high-quality decisions is more important than chasing short-term profits.

Profit is considered an outcome of disciplined analysis, robust risk management, and continuous improvement—not the primary objective itself.

Long-Term Vision

Although Version 1 focuses on the Forex market, AEGIS is designed to expand into:

- Stocks
- Cryptocurrencies
- Commodities
- Indices
- Bonds
- Exchange Traded Funds (ETFs)
- Futures
- Options (future phase)

This expansion should require minimal architectural changes due to the platform's modular design.

Strategic Objectives

Project AEGIS aims to:

- Build an institutional-grade financial intelligence platform.
- Develop a modular AI Council composed of specialized experts.
- Ensure every decision is fully explainable.
- Maintain permanent research records.
- Create a continuously evolving strategy ecosystem.
- Preserve every experiment and lesson learned.
- Eliminate emotional decision-making.
- Support future expansion into additional financial markets.
- Build a proprietary financial knowledge base.
- Operate with engineering discipline comparable to professional technology organizations.

Engineering Principles

Every component of AEGIS shall follow these principles:

1. Modularity

Each component performs one responsibility exceptionally well.

2. Explainability

Every recommendation must include:

- Supporting evidence.
- Confidence score.
- Reasoning.
- Contributing factors.

3. Scientific Thinking

- Every improvement begins with a hypothesis.
- Every hypothesis requires evidence.
- Every conclusion must be reproducible.

4. Continuous Learning

- Every trade becomes research.
- Every experiment contributes knowledge.
- Every failure improves future understanding.

5. Risk Before Opportunity

- Risk management has absolute authority.
- No AI system may bypass the Risk Engine.

6. Version Everything

The following assets shall always be version controlled:

- AI Models
- Strategies
- Databases
- APIs
- Documentation
- Risk Policies
- Feature Engineering Pipelines
- Experiments

7. Preserve Knowledge

Historical information should never be discarded without a documented reason. Institutional knowledge becomes increasingly valuable over time.

Long-Term Success Criteria

Project AEGIS will be considered successful when it demonstrates:

- Stable system operation.
- High-quality market intelligence.
- Explainable AI reasoning.
- Reliable risk management.
- Accurate research and experimentation.
- Continuous adaptation to market evolution.
- Scalable architecture.
- Comprehensive institutional knowledge.

Financial profitability is an important measure but shall never be the sole indicator of success.

Official Project Motto

Observe. Understand. Reason. Protect. Execute. Learn. Improve.

Closing Statement

Project AEGIS represents the vision of building an Artificial Financial Intelligence Platform founded upon engineering excellence, scientific methodology, explainable artificial intelligence, disciplined risk management, and continuous research.

Every architectural decision, implementation choice, and future enhancement should reinforce these principles so that AEGIS evolves into a trusted financial intelligence ecosystem capable of adapting alongside global financial markets for many years to come.

GUIDING PRINCIPLES

Document ID: AEGIS-DOC-002

Version: 1.0

Status: Approved

Classification: Core Engineering Documentation

Project: AEGIS – Artificial Financial Intelligence Platform

Executive Summary

This document establishes the engineering, scientific, architectural, and operational principles that govern the design, implementation, deployment, and evolution of Project AEGIS.

Every architectural decision, software component, AI model, research activity, and operational procedure shall conform to these principles unless formally revised through the Architecture Decision Record (ADR) process.

These principles act as the constitutional framework of AEGIS and are intended to preserve consistency as the platform grows over many years.

1. Engineering Philosophy

Project AEGIS shall always prioritize:

- Correctness over speed.
- Reliability over complexity.
- Simplicity over unnecessary sophistication.
- Long-term maintainability over short-term convenience.
- Scientific evidence over intuition.

Every feature must contribute measurable value to the platform.

2. Scientific Method

AEGIS shall operate according to the scientific method. Every improvement follows the cycle:

Observation → Question → Hypothesis → Experiment → Evaluation → Evidence → Conclusion → Knowledge → Improvement

No strategy shall enter production without evidence supporting its effectiveness.

3. Explainable Intelligence

Every recommendation produced by AEGIS shall answer the following questions:

- What decision was made?

- Why was it made?
- Which AI experts contributed?
- What evidence supported the decision?
- What confidence level was assigned?
- Which risks were identified?
- Why were alternative decisions rejected?

No unexplained recommendation shall reach the execution stage.

4. Risk Governance

Risk management is the highest authority within the platform. No system component, AI model, strategy, administrator, or future feature may bypass the Risk Engine.

Risk protection always takes precedence over profit generation. Capital preservation is considered a primary engineering objective.

5. Continuous Learning

AEGIS shall continuously improve through research. Every completed trade contributes to:

- AI evaluation.
- Strategy evaluation.
- Market understanding.
- Knowledge accumulation.
- Future experimentation.

Successes and failures shall receive equal analytical attention.

6. Modular Architecture

Each system component shall perform one clearly defined responsibility. Examples include:

- Market Data Service
- News Intelligence Service
- AI Council
- Risk Engine
- Execution Engine
- Research Laboratory
- Replay Engine
- Knowledge Graph

Components communicate through defined interfaces rather than direct internal dependencies.

7. Evidence-Based Decision Making

No recommendation shall rely solely upon:

- Personal opinion.
- Single indicators.
- Market rumors.
- Social media sentiment.
- Unverified information.

Every decision must be supported by measurable evidence obtained from validated data sources.

8. Reproducibility

Given identical market data, identical AI model versions, identical configurations, and identical strategy versions, AEGIS shall reproduce identical analytical results.

Reproducibility is essential for research integrity and debugging.

9. Version Control

The following assets shall always be version controlled:

- AI Models.
- Trading Strategies.
- Feature Engineering Pipelines.
- Database Schemas.
- API Specifications.
- Configuration Files.
- Documentation.
- Risk Policies.
- Experiments.

No production artifact shall exist without version identification.

10. Knowledge Preservation

Knowledge is one of AEGIS's most valuable assets. Historical information shall be preserved whenever practical. This includes:

- Trades.
- Experiments.
- AI reasoning.
- Market snapshots.
- Research findings.
- Strategy evolution.
- Performance reports.

Institutional knowledge compounds over time and should never be discarded without justification.

11. Scalability

Every architectural decision shall consider future expansion. The platform should be capable of supporting:

- Additional AI experts.
- Additional brokers.
- Additional exchanges.
- Additional asset classes.
- Larger datasets.
- Increased computational demand.

Scalability shall be achieved through modular design rather than major redesign.

12. Security

Security shall be incorporated from the beginning rather than added later. The platform shall implement:

- Authentication.
- Authorization.
- Encryption.
- Audit logging.
- Secure API communication.
- Secrets management.
- Principle of least privilege.

Security is considered an engineering requirement rather than an optional feature.

13. Observability

Every critical operation shall be observable. The platform shall continuously monitor:

- System health.
- AI performance.
- Latency.
- Resource utilization.
- Errors.
- Broker connectivity.
- Database performance.
- Model inference times.

Observability enables rapid diagnosis and continuous optimization.

14. Adaptability

Financial markets evolve continuously. AEGIS shall therefore be designed to:

- Detect market drift.

- Identify changing volatility.
- Evaluate declining model performance.
- Recommend retraining when appropriate.
- Support evolving strategies.

Adaptability shall be achieved through research rather than uncontrolled self-modification.

15. Human Oversight

Although AEGIS is intended to automate analysis and execution, human oversight remains essential. Humans remain responsible for:

- Defining objectives.
- Approving production deployments.
- Reviewing research.
- Evaluating architectural changes.
- Managing strategic direction.

Automation enhances human capability rather than replacing engineering judgment.

16. Ethical Responsibility

AEGIS shall be developed responsibly. The platform shall:

- Respect legal and regulatory requirements.
- Avoid deceptive practices.
- Maintain transparent decision processes.
- Preserve research integrity.
- Protect sensitive information.

Engineering excellence includes ethical responsibility.

17. Long-Term Vision

Every feature added to AEGIS should answer one question: Will this make the platform more intelligent, more reliable, more explainable, or more adaptable over the next five years?

If the answer is no, the feature should be reconsidered.

Constitutional Principles

The following principles shall never be violated without a formal Architecture Decision Record (ADR):

- Risk Engine has final authority.
- Every decision must be explainable.
- Every experiment must be reproducible.
- Knowledge must be preserved.
- Architecture shall remain modular.

- Research drives improvement.
- Documentation evolves with the system.
- Engineering discipline takes precedence over rapid feature development.

Closing Statement

These Guiding Principles define the engineering culture of Project AEGIS. They are intended to ensure that every future decision—whether architectural, technical, or operational—aligns with the project's long-term vision of building an adaptive, explainable, and scientifically grounded Artificial Financial Intelligence Platform.

SYSTEM ARCHITECTURE

Document ID: AEGIS-DOC-003

Version: 1.0

Status: Approved

Classification: Core Engineering Documentation

Project: AEGIS – Artificial Financial Intelligence Platform

Executive Summary

This document defines the high-level architecture of Project AEGIS. The architecture describes the major system components, their responsibilities, communication pathways, and operational boundaries.

The primary objective is to ensure scalability, modularity, explainability, maintainability, fault tolerance, and continuous evolution.

Every subsystem within AEGIS shall operate independently while cooperating through clearly defined communication interfaces.

1. Architectural Philosophy

Project AEGIS shall be designed as a distributed intelligence platform rather than a monolithic trading application. Every major responsibility shall exist as an independent subsystem. The platform should support:

- Independent development.
- Independent testing.
- Independent deployment.
- Independent scaling.
- Independent replacement.

No subsystem should become a bottleneck for the entire platform.

2. High-Level System Overview

PROJECT AEGIS: User Dashboard → API Gateway Layer → Intelligence Layer (AI Council, Chief Strategist, Research Director) → Feature Engine / Strategy Engine / Risk Engine → Market Data / News / Economic Calendar / Knowledge Graph → Historical Database / AI Memory / Research Database → Execution Engine → Broker APIs.

3. Architectural Layers

The platform consists of eight logical layers.

Layer 1 — Presentation Layer

Purpose: Provide interfaces for users and administrators.

Responsibilities: Dashboard, Analytics, Charts, Monitoring, Reports, Research Viewer, Trade History, Configuration.

Technologies: React, TypeScript, Tailwind CSS, Framer Motion.

Layer 2 — API Gateway

Purpose: Single entry point for all clients.

Responsibilities: Authentication, Authorization, Rate Limiting, Request Validation, API Routing, Logging.

No business logic shall exist within the gateway.

Layer 3 — Intelligence Layer

This is the brain of AEGIS. Contains: AI Council, Chief Strategist, Research Director.

Responsibilities: Market reasoning, Expert collaboration, Decision generation, Continuous evaluation.

This layer never communicates directly with brokers.

Layer 4 — Decision Layer

Purpose: Transform intelligence into executable decisions. Contains: Strategy Engine, Feature Engine, Risk Engine.

Responsibilities: Strategy selection, Feature generation, Risk validation, Position sizing, Trade approval.

No trade may bypass this layer.

Layer 5 — Intelligence Sources

Purpose: Observe financial markets. Sources include: Live Market Data, Financial News, Economic Calendars, Central Bank Events, Knowledge Graph, Historical Data.

This layer never executes trades. Its responsibility is observation.

Layer 6 — Data Layer

Purpose: Store institutional knowledge. Contains: Market Database, Research Database, AI Memory, Strategy Repository, Knowledge Graph, Experiment Registry, Event Library.

All historical knowledge is preserved here.

Layer 7 — Execution Layer

Purpose: Execute approved trades.

Responsibilities: Broker Communication, Order Placement, Order Modification, Position Monitoring, Trade Closure, Error Recovery.

Execution does not make trading decisions. It only executes approved decisions.

Layer 8 — Operations Layer

Purpose: Maintain platform reliability.

Responsibilities: Monitoring, Logging, Metrics, Health Checks, Backups, Deployment, Alerting, Security Auditing.

This layer ensures operational excellence.

4. Major System Components

The platform consists of the following primary subsystems:

Intelligence Division

- AI Council
- Chief Strategist
- Research Director

Market Intelligence Division

- Market Data Collector
- News Intelligence
- Economic Calendar
- Sentiment Analyzer
- Correlation Analyzer

Decision Division

- Strategy Engine
- Feature Engineering
- Decision Pipeline
- Risk Engine

Trading Division

- Execution Engine
- Broker Connectors
- Position Manager
- Portfolio Manager

Research Division

- Backtesting Engine
- Paper Trading
- Replay Engine
- Experiment Laboratory

Knowledge Division

- Knowledge Graph
- Event Library

- Pattern Library
- Strategy Library
- AI Memory

Operations Division

- Monitoring
- Deployment
- Security
- Configuration
- Logging
- Metrics

5. Data Flow Philosophy

Every market event follows a structured pipeline: Market Observation → Validation → Feature Engineering → AI Council Analysis → Chief Strategist Review → Risk Validation → Execution Approval → Trade Execution → Outcome Recording → Research → Knowledge Update → Continuous Learning.

No stage shall be skipped.

6. System Characteristics

AEGIS shall be:

- Modular — Every subsystem has one responsibility.
- Explainable — Every decision can be reconstructed.
- Fault Tolerant — Failure of one component should not collapse the platform.
- Scalable — Additional AI experts or services can be integrated without redesign.
- Observable — Every critical action is logged and measurable.
- Adaptable — The platform evolves through research rather than hard-coded assumptions.

7. Architectural Constraints

The following constraints shall always apply:

- Risk Engine possesses final authority.
- AI systems cannot directly communicate with brokers.
- Execution Engine cannot create trading ideas.
- Research systems cannot alter production data directly.
- Historical records remain immutable.
- Every production change must be version controlled.

8. Future Expansion

The architecture shall support future integration of:

- Stocks
- Cryptocurrency
- Commodities
- Options
- Futures
- Institutional Order Flow
- Reinforcement Learning
- Multi-Broker Routing
- Distributed Computing
- GPU Acceleration
- Cloud-Native Scaling

No fundamental redesign should be required.

9. Architectural Principles

Every future subsystem must satisfy: Single Responsibility, Loose Coupling, High Cohesion, Clear Interfaces, Version Compatibility, Explainability, Testability, Security, Scalability.

10. Closing Statement

The System Architecture establishes the structural foundation of Project AEGIS. It defines the major layers, system boundaries, responsibilities, and communication pathways that enable the platform to evolve into an institutional-grade Artificial Financial Intelligence Platform.

Every future engineering decision shall align with this architecture unless formally revised through the Architecture Decision Record process.

COGNITIVE ARCHITECTURE

Document ID: AEGIS-DOC-004

Version: 1.0

Status: Approved

Classification: Core Engineering Documentation

Project: AEGIS – Artificial Financial Intelligence Platform

Executive Summary

This document defines the Cognitive Architecture of Project AEGIS. The Cognitive Architecture describes how AEGIS observes financial markets, processes information, reasons about uncertainty, evaluates competing hypotheses, manages risk, learns from outcomes, and continuously improves its decision-making capabilities.

Unlike traditional algorithmic trading systems that rely on static rules, AEGIS is designed to emulate a disciplined financial research organization composed of specialized experts working collaboratively.

The Cognitive Architecture is the intellectual foundation of the platform.

1. Cognitive Philosophy

AEGIS is designed around one principle: Thinking precedes action.

Every trade begins with understanding. Every understanding begins with observation. Every observation must be validated before reasoning.

The objective is not to predict markets with certainty, but to make the highest-quality decision possible given the available evidence.

2. The Cognitive Cycle

Every market opportunity follows the same reasoning cycle: Observe → Validate → Understand → Generate Features → Reason → Debate → Evaluate Risk → Decide → Execute → Observe Outcome → Learn → Improve.

This cycle forms the foundation of all intelligence within AEGIS.

3. Stage One — Observation

Purpose: Collect information from multiple independent sources.

Observation includes: Live market prices, Candlestick data, Technical indicators, Economic calendar events, Financial news, Central bank announcements, Historical market behavior, Volatility measurements, Liquidity conditions, Correlation data.

Observation does not include interpretation. Its only objective is accurate data collection.

4. Stage Two — Validation

Every observation must be validated before it enters the reasoning process. Validation checks include: Missing market data, Duplicate ticks, Invalid prices, Outlier detection, API reliability, Timestamp consistency, News authenticity, Data freshness.

Only validated information proceeds further.

5. Stage Three — Understanding

Understanding transforms raw observations into meaningful context. Examples include: Market regime identification, Trend classification, Session recognition, Volatility assessment, Liquidity estimation, Sentiment interpretation, Macroeconomic context, Intermarket relationships.

At this stage, AEGIS begins to understand the environment rather than merely recording it.

6. Stage Four — Feature Generation

Validated information is transformed into structured analytical features. Examples: Moving averages, Relative Strength Index (RSI), Average True Range (ATR), Trend strength, Sentiment score, Economic impact score, Correlation coefficients, Market regime classification, Liquidity index, Session characteristics.

These features become the language understood by the AI Council.

7. Stage Five — Expert Reasoning

Each specialized AI expert independently evaluates the same market snapshot. Experts do not influence each other directly.

Each expert produces: Recommendation, Confidence score, Supporting evidence, Explanation, Risk observations.

Independent reasoning reduces confirmation bias.

8. Stage Six — Collaborative Debate

The AI Council submits all expert opinions to the Chief Strategist. The Chief Strategist evaluates: Areas of agreement, Areas of disagreement, Historical performance of each expert, Current market conditions, Reliability of evidence.

The objective is not majority voting but evidence-weighted reasoning.

9. Stage Seven — Risk Evaluation

Before any decision proceeds further, the Risk Engine performs an independent review. Risk evaluation includes: Position sizing, Maximum exposure, Drawdown limits, Correlated positions, Portfolio impact, Expected volatility, Market conditions, Capital preservation.

The Risk Engine possesses final approval authority.

10. Stage Eight — Decision Formation

The Chief Strategist prepares a decision proposal. Every proposal contains: Direction (Buy, Sell, Hold), Confidence score, Supporting evidence, AI contributors, Risk assessment, Recommended position size, Expected holding period.

No proposal is executed automatically.

11. Stage Nine — Execution

The Execution Engine receives only approved decisions. Responsibilities include: Broker communication, Order submission, Order confirmation, Position monitoring, Error recovery, Execution logging.

Execution contains no analytical logic.

12. Stage Ten — Outcome Analysis

After every completed trade, AEGIS performs a structured review. Questions include: Was the prediction correct? Which experts contributed most? Which assumptions failed? Which evidence proved valuable? Were risks underestimated? How did execution quality affect results?

Every outcome becomes research.

13. Stage Eleven — Learning

Learning updates institutional knowledge. Examples: AI performance statistics, Strategy effectiveness, Market behavior changes, Confidence calibration, Feature importance, Knowledge Graph expansion, Research findings.

Learning influences future versions rather than modifying live production behavior without validation.

14. Cognitive Memory

AEGIS maintains several forms of memory.

Short-Term Memory

Current market state. Current open positions. Current news. Current volatility.

Long-Term Memory

Historical trades. Research findings. AI performance. Market events. Strategies. Experiments. Knowledge Graph.

Episodic Memory

Individual trading events. Every trade records: Market snapshot, AI reasoning, Decision process, Execution, Final outcome.

Semantic Memory

General financial knowledge. Examples: Central banks, Economic indicators, Market relationships, Trading principles, Risk policies.

15. Cognitive Objectives

The intelligence system must continuously strive to: Improve reasoning quality, Reduce uncertainty, Detect changing market conditions, Increase explainability, Preserve institutional knowledge, Enhance research quality, Protect trading capital.

16. Cognitive Constraints

The following rules shall never be violated:

- Raw data shall never bypass validation.
- AI experts shall reason independently.
- Risk approval is mandatory.
- Learning shall never overwrite historical evidence.
- Production behavior changes require validation through research.

17. Future Cognitive Expansion

The Cognitive Architecture shall support future capabilities including: Reinforcement Learning, Multi-Agent Collaboration, Long-Term Strategic Planning, Cross-Market Intelligence, Institutional Flow Analysis, Autonomous Research Generation, Adaptive Expert Weighting, Explainable Deep Learning.

These enhancements shall integrate without requiring redesign of the core cognitive framework.

Closing Statement

The Cognitive Architecture establishes how AEGIS transforms information into knowledge and knowledge into disciplined action. By separating observation, reasoning, collaboration, risk governance, execution, and learning into distinct cognitive stages, the platform achieves explainability, adaptability, and scientific rigor.

This architecture ensures that every trading decision is the result of structured intelligence rather than isolated algorithms, providing a foundation for continuous evolution as financial markets and AI technologies advance.

AI COUNCIL ARCHITECTURE

Document ID: AEGIS-DOC-005

Version: 1.0

Status: Approved

Classification: Core Engineering Documentation

Project: AEGIS – Artificial Financial Intelligence Platform

Executive Summary

This document defines the Artificial Intelligence Council, the central intelligence organization responsible for market analysis, collaborative reasoning, evidence evaluation, strategic recommendation generation, and continuous learning.

Rather than relying on a single artificial intelligence model, AEGIS adopts a multi-agent architecture composed of specialized AI experts. Each expert focuses on a specific aspect of market analysis and independently evaluates every trading opportunity before contributing its findings to the Chief Strategist.

This architecture is designed to maximize explainability, modularity, adaptability, and long-term maintainability.

1. Design Philosophy

AEGIS treats intelligence as an organization rather than a single algorithm. The AI Council functions similarly to an institutional investment committee.

Each expert specializes in one domain. The Chief Strategist integrates evidence from all experts. The Risk Engine independently validates every recommendation.

This separation reduces bias and improves decision quality.

2. AI Organizational Structure

Research Director → Chief Strategist AI → [Trend AI, News AI, Pattern AI, Liquidity AI, Macro AI, Session AI, Correlation AI, Risk AI (Advisory)] → Decision Recommendation → Risk Engine → Execution Engine.

3. AI Council Objectives

The AI Council shall: Analyze every market opportunity, Provide independent expert opinions, Reduce analytical bias, Produce explainable recommendations, Continuously evaluate expert performance, Adapt to changing market conditions, Preserve institutional knowledge.

4. Core Design Principles

Every AI expert shall satisfy the following principles: Single Responsibility, Explainability, Measurable Performance, Independent Reasoning, Version Control, Continuous Evaluation, Modular Integration.

Experts shall never directly influence each other's reasoning before the Chief Strategist review.

5. Trend Intelligence AI

Mission: Determine the dominant market trend.

Responsibilities: Trend identification, Trend strength measurement, Momentum analysis, Trend continuation assessment, Trend exhaustion detection.

Primary Inputs: EMA, SMA, ADX, ATR, MACD, RSI, Price Structure, Higher Highs / Lower Lows.

Outputs: Market Direction, Trend Strength, Confidence Score, Supporting Evidence, Explanation.

Strengths: Trending markets, Momentum continuation. Weaknesses: Sideways markets, False breakouts.

6. News Intelligence AI

Mission: Understand financial news and economic developments.

Responsibilities: News classification, Sentiment analysis, Impact estimation, Event duration estimation, Currency impact assessment.

Data Sources: Financial news providers, Central bank announcements, Economic calendars, Government releases.

Outputs: Currency Sentiment, Impact Level, Confidence Score, Expected Duration, Supporting Evidence.

7. Pattern Intelligence AI

Mission: Recognize technical market structures.

Responsibilities: Detect patterns including Double Top, Double Bottom, Head and Shoulders, Flags, Pennants, Wedges, Channels, Triangles, Cup and Handle. Future support may include Smart Money Concepts (SMC) after research validation.

Outputs: Pattern Type, Breakout Probability, Pattern Reliability, Confidence Score.

8. Liquidity Intelligence AI

Mission: Estimate market liquidity and identify probable liquidity targets.

Responsibilities: Stop-loss clustering, Liquidity sweeps, Fake breakout detection, Order imbalance estimation, Liquidity zones.

Outputs: Liquidity Map, Liquidity Risk, Confidence Score, Trade Opportunity Assessment.

9. Macroeconomic Intelligence AI

Mission: Evaluate long-term economic conditions.

Inputs: Interest Rates, Inflation, GDP, Employment, Bond Yields, Central Bank Policy.

Outputs: Currency Bias, Economic Trend, Long-Term Confidence, Supporting Evidence.

10. Session Intelligence AI

Mission: Understand market session behavior.

Responsibilities: Session identification, Historical session analysis, Expected volatility, Liquidity estimation, Session-specific behavior.

Sessions: Sydney, Tokyo, London, New York, Session Overlaps.

11. Correlation Intelligence AI

Mission: Analyze relationships between financial instruments.

Examples: EUR/USD, GBP/USD, USD/JPY, Gold, DXY, Bond Markets.

Responsibilities: Correlation analysis, Divergence detection, Cross-market confirmation.

12. Risk Intelligence AI (Advisory)

Mission: Provide analytical risk recommendations.

Responsibilities: Volatility assessment, Drawdown estimation, Exposure estimation, Suggested position sizing.

Important: Risk AI provides advice only. Final approval belongs exclusively to the Risk Engine.

13. Chief Strategist AI

The Chief Strategist coordinates the AI Council. Responsibilities include: Reviewing all expert opinions, Identifying agreement and disagreement, Measuring evidence quality, Weighting experts based on historical performance, Producing a unified recommendation, Explaining the final reasoning.

The Chief Strategist never communicates directly with broker APIs.

14. Research Director

The Research Director is responsible for long-term intelligence improvement. Responsibilities include: Monitoring AI performance, Detecting model drift, Evaluating strategy evolution, Identifying research opportunities, Recommending retraining, Supervising experiments.

The Research Director never participates in live trading decisions.

15. Expert Performance Evaluation

Every expert shall maintain historical metrics including: Accuracy, Precision, Recall, Confidence Calibration, Average Latency, False Positives, False Negatives, Market-Specific Performance.

Historical performance influences future weighting.

16. Dynamic Expert Weighting

Expert influence shall vary according to market conditions. Examples: During major economic announcements, News AI and Macro AI receive greater weighting. During stable technical markets, Trend AI and Pattern AI receive greater weighting.

Weight adjustments shall be evidence-based and continuously evaluated.

17. Decision Generation

Every recommendation shall include: Buy / Sell / Hold, Confidence Score, Supporting Experts, Evidence Summary, Risk Assessment, Recommended Position Size, Expected Holding Horizon.

18. AI Memory

Every AI expert maintains historical memory. Memory includes: Previous predictions, Confidence history, Accuracy history, Strengths, Weaknesses, Learning history, Model versions.

This enables continuous performance evaluation.

19. Future AI Expansion

Future experts may include: Institutional Flow AI, Options Intelligence AI, Volatility Forecast AI, Reinforcement Learning Agent, Order Book Intelligence AI, Cross-Asset Intelligence AI, Seasonal Intelligence AI.

The architecture supports expansion without redesign.

20. Constitutional Rules

The AI Council shall always comply with the following principles:

- Experts reason independently.
- No expert may directly execute trades.
- Every recommendation must be explainable.
- Risk Engine retains final authority.
- All recommendations are permanently recorded.
- Model versions shall always be tracked.
- Historical reasoning shall remain reproducible.

Closing Statement

The AI Council Architecture establishes the intelligence organization at the core of Project AEGIS. By distributing analytical responsibilities across specialized experts, coordinating evidence through the Chief Strategist, and separating strategic reasoning from risk governance, AEGIS achieves transparency, adaptability, and institutional-grade decision quality.

This architecture ensures that intelligence evolves through research, measurement, and continuous improvement while maintaining explainability and engineering discipline.

DATA FLOW & DECISION PIPELINE

Document ID: AEGIS-DOC-006

Version: 1.0

Status: Approved

Classification: Core Engineering Documentation

Project: AEGIS – Artificial Financial Intelligence Platform

Executive Summary

This document specifies the end-to-end flow of information within Project AEGIS. It defines how market observations are transformed into structured intelligence, how decisions are produced through collaborative reasoning, how risk is validated, how trades are executed, and how every outcome contributes to continuous learning.

Every decision follows a deterministic, traceable, and reproducible pipeline.

1. Pipeline Philosophy

AEGIS follows a strict engineering principle: No information shall bypass the decision pipeline. Every trading decision must pass through each stage in sequence.

This guarantees: Explainability, Traceability, Risk control, Reproducibility, Scientific integrity.

2. High-Level Pipeline

Market Observation → Data Validation → Feature Engineering → Market Snapshot Creation → AI Council Analysis → Chief Strategist Review → Risk Engine Validation → Execution Engine → Broker API → Trade Monitoring → Trade Closure → Outcome Analysis → Research Database → Knowledge Graph Update.

3. Stage One – Market Observation

The Market Intelligence Layer continuously collects data. Sources include: Broker market feeds, Tick data, OHLC candles, Economic calendars, Financial news, Central bank announcements, Market session information, Historical databases.

No interpretation occurs at this stage. Only observation.

4. Stage Two – Data Validation

Incoming data is validated before entering the platform. Validation checks include: Missing values, Invalid timestamps, Duplicate records, Out-of-range prices, API failures, Corrupted payloads, Delayed market feeds.

Invalid data is quarantined and logged.

5. Stage Three – Feature Engineering

Validated data is transformed into analytical features.

Technical Features: RSI, MACD, EMA, SMA, ATR, Bollinger Bands.

Market Features: Trend Strength, Volatility Index, Liquidity Score, Session Classification, Correlation Matrix.

Economic Features: News Sentiment, Economic Impact Score, Interest Rate Differential.

6. Stage Four – Market Snapshot

AEGIS creates a complete market snapshot. A snapshot includes: Timestamp, Currency Pair, Current Price, Technical Indicators, Market Regime, Session, Volatility, Liquidity, Correlation, News Context, Economic Calendar, Feature Set Version.

This snapshot becomes the reference point for all reasoning.

7. Stage Five – AI Council Analysis

Each AI expert independently analyzes the market snapshot. Outputs include: Recommendation, Confidence, Evidence, Explanation, Risk Observations.

Experts do not communicate directly with one another.

8. Stage Six – Chief Strategist Review

The Chief Strategist receives all expert analyses. Responsibilities: Detect agreement, Detect disagreement, Evaluate evidence quality, Apply dynamic expert weighting, Produce unified recommendation, Generate reasoning summary.

Outputs include: Decision, Confidence, Supporting experts, Evidence summary, Strategic explanation.

9. Stage Seven – Risk Engine

Before execution, every recommendation is evaluated independently. Checks include: Position sizing, Portfolio exposure, Maximum drawdown, Correlation risk, Volatility limits, Capital allocation, Daily loss limits.

Possible outcomes: Approved, Approved with reduced position size, Deferred, Rejected.

No recommendation bypasses this stage.

10. Stage Eight – Execution Engine

Approved decisions are transmitted to the Execution Engine. Responsibilities include: Broker authentication, Order placement, Stop-loss placement, Take-profit placement, Order confirmation, Error handling, Retry logic, Slippage recording.

Execution does not modify trading decisions.

11. Stage Nine – Position Monitoring

Open trades are monitored continuously. Monitoring includes: Unrealized profit/loss, Stop-loss status, Take-profit status, Margin utilization, Market volatility, Broker connectivity.

Important events are recorded.

12. Stage Ten – Trade Closure

Trades close through: Take Profit, Stop Loss, Manual intervention, Risk protection, Time expiration, Strategy exit conditions.

Closure information is permanently stored.

13. Stage Eleven – Outcome Analysis

After closure, AEGIS evaluates: Prediction accuracy, AI expert contributions, Strategy effectiveness, Risk performance, Execution quality, Slippage impact, Market behavior.

Every trade becomes research.

14. Stage Twelve – Knowledge Update

Results are written to: Research Database, AI Memory, Strategy Library, Experiment Registry, Knowledge Graph, Decision Archive.

Nothing is discarded.

15. Event-Driven Communication

Every stage publishes events. Examples: MarketSnapshotCreated, FeatureGenerationCompleted, AIAnalysisCompleted, DecisionGenerated, RiskApproved, RiskRejected, TradeExecuted, TradeClosed, KnowledgeUpdated.

Services subscribe only to relevant events. This enables loose coupling and scalability.

16. Failure Recovery

If any stage fails: The pipeline stops safely, the failure is logged, an alert is generated, recovery procedures begin. No partial trade execution occurs.

Safety always takes precedence over availability.

17. Observability

Every stage records: Start time, End time, Processing duration, Resource utilization, Input references, Output references, Errors, Version identifiers.

This enables complete auditing.

18. Pipeline Performance Objectives

Target metrics: Data validation latency: <100 ms, Feature generation: <250 ms, AI Council analysis: <500 ms, Risk evaluation: <100 ms, Execution request preparation: <100 ms.

These targets may be refined as the platform evolves.

19. Constitutional Rules

The pipeline shall always satisfy:

- No skipped stages.
- Every decision is traceable.
- Every recommendation is explainable.
- Every trade is auditable.
- Every outcome contributes to research.
- Every failure is recoverable.
- Historical data remains immutable.

Closing Statement

The Data Flow & Decision Pipeline defines the operational heartbeat of Project AEGIS. By enforcing a structured sequence from observation to learning, the platform ensures that every trading decision is transparent, reproducible, and governed by evidence rather than intuition.

This pipeline establishes the operational discipline required for a long-lived Artificial Financial Intelligence Platform capable of adapting through continuous research and institutional knowledge accumulation.

DATABASE ARCHITECTURE

Document ID: AEGIS-DOC-007

Version: 1.0

Status: Approved

Classification: Core Engineering Documentation

Project: AEGIS – Artificial Financial Intelligence Platform

Executive Summary

The Database Architecture defines how Project AEGIS stores, organizes, retrieves, protects, and evolves information throughout its operational lifecycle.

Unlike conventional trading systems that primarily record executed trades, AEGIS treats data as institutional knowledge. Every observation, market event, AI reasoning process, experiment, strategy revision, and execution outcome becomes part of the platform's long-term memory.

The architecture is designed to ensure: Scalability, High availability, Data integrity, Auditability, Explainability, Continuous learning, Efficient analytics, Research reproducibility.

1. Database Philosophy

The database is not merely a storage system. It is the memory of AEGIS.

Every decision made by the platform must be explainable years later using stored information. Therefore, data is considered one of the platform's most valuable assets.

2. Design Principles

The database shall follow these principles: Single Source of Truth, ACID Compliance, Immutable Historical Records, Normalized Operational Data, Optimized Analytical Storage, Versioned Knowledge, Event Traceability, Strong Referential Integrity, Security by Design.

3. High-Level Database Architecture

AEGIS DATABASE: Configuration Store → Market Data Database / Research Database / Knowledge Graph Database / Strategy Repository / AI Memory Database / Execution Database / Monitoring Database / Experiment Database / Audit Database → Analytics Warehouse.

Each database has a distinct responsibility.

4. Core Databases

4.1 Market Data Database

Purpose: Store all historical and real-time market information.

Contents: Tick Data, OHLC Candles, Volume, Spread, Volatility, Session Information, Liquidity Metrics.

Retention: Permanent. Historical market data is never deleted.

4.2 Research Database

Purpose: Store all research activities.

Contains: Experiments, Backtests, Research Notes, Hypotheses, Conclusions, Statistical Reports.

Every research activity receives a unique identifier.

4.3 AI Memory Database

Purpose: Maintain historical AI behavior.

Stores: Predictions, Confidence Scores, Reasoning, Feature Importance, Model Versions, Performance History, Accuracy Statistics.

This allows future evaluation of every AI model.

4.4 Strategy Repository

Purpose: Version control for trading strategies.

Each strategy records: Strategy ID, Version, Parameters, Performance, Deployment Date, Retirement Date, Validation Results.

Strategies are never overwritten. Every revision creates a new version.

4.5 Execution Database

Purpose: Maintain complete trading records.

Stores: Orders, Executions, Position History, Slippage, Broker Responses, Execution Errors.

Every execution receives a permanent audit trail.

4.6 Knowledge Graph

Purpose: Represent relationships. Examples: Currency Relationships, Economic Indicators, Market Events, AI Experts, Trading Strategies, Historical Patterns.

Unlike relational tables, the Knowledge Graph captures connections between entities.

4.7 Experiment Database

Purpose: Track scientific experimentation.

Stores: Hypothesis, Dataset, Parameters, Results, Statistical Significance, Conclusions, Reviewer, Approval Status.

Every experiment remains reproducible.

4.8 Monitoring Database

Purpose: System health.

Stores: CPU Usage, Memory Usage, API Latency, AI Inference Time, Queue Length, Error Logs, Broker Latency.

Supports operational excellence.

4.9 Audit Database

Purpose: Security and compliance.

Stores: Authentication Events, Configuration Changes, Strategy Deployments, AI Model Updates, User Actions, Administrative Activities.

Audit records are immutable.

5. Entity Relationships

Major entities include: Markets, Instruments, Market Snapshots, AI Models, AI Decisions, Strategies, Trades, Risk Evaluations, Research Projects, Experiments, Knowledge Nodes, Broker Sessions, Portfolio Positions.

Relationships shall be explicitly defined and version controlled.

6. Data Lifecycle

Every record follows the lifecycle: Created → Validated → Stored → Referenced → Analyzed → Archived → Retained.

Deletion should be avoided unless legally or operationally required.

7. Versioning

The following assets must always include version metadata: AI Models, Trading Strategies, Feature Pipelines, Database Schemas, Research Papers, Knowledge Graph, APIs.

Version history shall remain permanently accessible.

8. Data Integrity

The platform shall enforce: Primary Keys, Foreign Keys, Unique Constraints, Check Constraints, Transaction Management, ACID Properties.

Consistency is mandatory.

9. Security

Security requirements include: Encryption at Rest, Encryption in Transit, Role-Based Access Control (RBAC), Principle of Least Privilege, Secret Management, Audit Logging, Regular Backups.

Sensitive information shall never be stored in plaintext.

10. Backup Strategy

Backups include: Daily Incremental, Weekly Full Backup, Monthly Archive, Annual Snapshot.

Recovery procedures shall be tested regularly.

11. Data Retention Policy

Permanent: Trades, AI Decisions, Research, Experiments, Knowledge Graph, Strategy History.

Temporary: Cached Data, Session Tokens, Temporary Queues.

Retention policies shall balance historical value with operational efficiency.

12. Analytics Warehouse

A dedicated analytical environment shall aggregate data for: Performance Reporting, AI Evaluation, Strategy Comparison, Long-Term Trends, Executive Dashboards.

Operational databases should not be overloaded with analytical workloads.

13. Scalability

The database architecture shall support: Horizontal Scaling, Read Replicas, Partitioning, Compression, Archiving, Distributed Storage.

Future growth should not require redesign.

14. Constitutional Rules

The database architecture shall always satisfy:

- Historical records are immutable.
- Every trade is traceable.
- Every AI decision is explainable.
- Every experiment is reproducible.
- Every strategy is versioned.
- Every configuration change is audited.
- Data integrity takes precedence over convenience.

15. Future Expansion

Future versions may introduce: Vector Database for semantic AI memory, Time-Series Database optimized for tick data, Graph Database dedicated to the Knowledge Graph, Data Lake for raw market archives, Distributed analytical clusters.

The architecture is intentionally modular to accommodate these additions.

Closing Statement

The Database Architecture establishes the institutional memory of Project AEGIS. By treating data as a strategic asset rather than passive storage, the platform ensures that every observation, decision, experiment, and outcome contributes to a growing body of financial intelligence.

This design enables AEGIS to remain explainable, auditable, and continuously improving throughout its operational lifetime.

TECHNOLOGY STACK & INFRASTRUCTURE

Document ID: AEGIS-DOC-008

Version: 1.0

Status: Approved

Classification: Core Engineering Documentation

Project: AEGIS – Artificial Financial Intelligence Platform

Executive Summary

This document defines the official technology stack for Project AEGIS. The stack has been selected based on five primary objectives: High performance, Scalability, Reliability, Cost efficiency, Long-term maintainability.

Version 1 of AEGIS is designed to leverage open-source technologies wherever practical while taking advantage of cloud infrastructure available through the GitHub Student Developer Pack, including DigitalOcean credits.

1. Technology Selection Principles

Every technology introduced into AEGIS must satisfy the following requirements: Production-ready, Actively maintained, Strong community support, Well documented, Horizontally scalable where applicable, Compatible with Python-first development, Suitable for financial and AI workloads.

Technology choices should prioritize stability over novelty.

2. High-Level Technology Architecture

USER → React + TypeScript Dashboard → FastAPI Backend → [Redis, PostgreSQL + TimescaleDB, Kafka/NATS, Object Storage, MLflow, Research Database] → [AI Models, Feature Engineering, AI Council, Risk Engine, Execution Engine] → Docker Containers → DigitalOcean Cloud → Monitoring & Logging.

3. Frontend

Purpose: Provide dashboards, analytics, configuration, and operational visibility.

Official Stack: React, TypeScript, Vite, Tailwind CSS, Framer Motion, TanStack Query, React Router.

Responsibilities: Live market visualization, AI explanations, Trade monitoring, Research dashboards, System administration.

4. Backend

Purpose: Business logic and orchestration.

Official Stack: Python, FastAPI, Pydantic, SQLAlchemy, Alembic.

Responsibilities: API services, Authentication, AI orchestration, Feature engineering, Risk management, Broker communication.

Python is selected due to its exceptional AI and quantitative finance ecosystem.

5. Databases

Primary Database: PostgreSQL — Purpose: Core relational data, User data, Trades, Research, Configuration.

Time-Series Database Extension: TimescaleDB — Purpose: Tick data, Candlestick history, High-frequency market observations, Performance metrics.

6. Caching

Technology: Redis.

Responsibilities: Market cache, Session cache, AI intermediate results, Rate limiting, Distributed locks, Task coordination.

Redis significantly reduces database load and improves response times.

7. Messaging

Preferred Technology: NATS (initial deployment). Future Option: Apache Kafka.

Purpose: Enable asynchronous communication between independent services, such as market data ingestion, AI analysis, and execution.

NATS is lightweight and ideal for Version 1. Kafka can be introduced if message volume grows substantially.

8. AI & Machine Learning

Programming Language: Python.

Libraries: PyTorch, XGBoost, LightGBM, Scikit-learn, Pandas, NumPy, Polars.

Experiment Tracking: MLflow.

Responsibilities: Model versioning, Experiment tracking, Performance comparison, Reproducibility.

9. Workflow Orchestration

Technology: Prefect.

Purpose: Scheduled data ingestion, Research pipelines, Backtesting jobs, Model retraining, Report generation.

Prefect is chosen for its modern developer experience and flexibility.

10. Containerization

Technology: Docker.

Purpose: Consistent development environments, Reproducible deployments, Service isolation, Simplified scaling.

Every major AEGIS service should run within its own container.

11. Cloud Infrastructure

Primary Provider: DigitalOcean.

Reason: GitHub Student Developer Pack credits, Developer-friendly platform, Affordable pricing, Managed databases, Object storage, Kubernetes support for future scaling.

Version 1 Deployment: Docker containers, DigitalOcean Droplet, Managed PostgreSQL, Spaces (Object Storage).

12. Monitoring & Observability

Technologies: Prometheus, Grafana, Loki, OpenTelemetry.

Responsibilities: Infrastructure monitoring, AI inference metrics, API latency, Database performance, System health, Alerting.

13. Security

Authentication: JWT + Refresh Tokens.

Secrets Management: Environment variables with secure secret storage.

Security Practices: HTTPS everywhere, Role-Based Access Control (RBAC), API rate limiting, Input validation, Audit logging, Database encryption, Secure broker credential storage.

14. Development Tools

Version Control: Git + GitHub.

Code Quality: Ruff, Black, isort, MyPy.

Testing: Pytest, Playwright (Frontend), HTTPX (API).

Documentation: Markdown + OpenAPI (Swagger).

15. Infrastructure Evolution

Version 1: Single DigitalOcean region, Docker, FastAPI, PostgreSQL, TimescaleDB, Redis, NATS.

Version 2: Multiple AI workers, Load balancing, Read replicas, Background processing cluster.

Version 3: Kubernetes, Auto-scaling, Multi-region deployment, Disaster recovery, High availability architecture.

16. Technology Governance

New technologies shall only be adopted if they: Solve a measurable problem, Integrate cleanly with the architecture, Improve reliability or scalability, Do not introduce unnecessary complexity.

Technology adoption should be driven by need, not trend.

Closing Statement

The AEGIS Technology Stack provides a modern, scalable, and cost-conscious foundation for building an institutional-grade Artificial Financial Intelligence Platform. By combining proven open-source technologies with cloud infrastructure supported by the GitHub Student Developer Pack, the platform is positioned for sustainable growth from initial development through production deployment and future expansion.

DEVELOPMENT ROADMAP

Document ID: AEGIS-DOC-009

Version: 1.0

Status: Approved

Classification: Core Engineering Documentation

Project: AEGIS – Artificial Financial Intelligence Platform

Executive Summary

This roadmap defines the official implementation strategy for Project AEGIS.

The development approach follows a principle of progressive evolution. Each phase builds upon the previous one, ensuring that the platform remains stable, testable, and scientifically validated before advancing.

The roadmap prioritizes reliability and evidence over rapid feature development.

Development Philosophy

The project shall be developed in incremental phases. Each phase must satisfy three conditions before progressing: Functional completion, Comprehensive testing, Documentation updates.

No phase shall be skipped.

Phase 0 — Project Foundation

Objective: Establish the engineering foundation for AEGIS.

Deliverables: GitHub repository, Project structure, Docker development environment, FastAPI backend, React frontend, PostgreSQL + TimescaleDB, Redis, Configuration management, Authentication, Logging framework, CI/CD pipeline, Initial deployment on DigitalOcean.

Success Criteria: Local development environment fully operational. Cloud deployment verified. Automated builds and tests functioning.

Phase 1 — Market Intelligence Layer

Objective: Collect and organize market information.

Deliverables: Broker API integration, Real-time market data ingestion, Historical data storage, Economic calendar integration, News aggregation, Session detection, Market validation pipeline.

Success Criteria: Reliable, validated market data available for analysis. No data loss during continuous operation.

Phase 2 — Feature Engineering Engine

Objective: Transform raw market data into structured analytical features.

Deliverables: Technical indicator generation, Volatility metrics, Liquidity metrics, Correlation calculations, Sentiment scoring, Market regime classification, Market snapshot generation.

Success Criteria: Consistent feature generation with version tracking. Reproducible outputs from identical inputs.

Phase 3 — AI Council

Objective: Develop the core intelligence system.

Deliverables: Trend AI, News AI, Pattern AI, Liquidity AI, Macro AI, Session AI, Correlation AI, Chief Strategist, AI explanation engine.

Success Criteria: Each AI expert produces independent recommendations. Chief Strategist generates explainable unified decisions.

Phase 4 — Risk Management

Objective: Protect trading capital.

Deliverables: Position sizing engine, Maximum drawdown controls, Exposure management, Correlation risk analysis, Daily loss limits, Portfolio protection, Trade approval workflow.

Success Criteria: Every trade passes through the Risk Engine. Risk policies are fully enforceable.

Phase 5 — Research Laboratory

Objective: Build the scientific foundation for continuous improvement.

Deliverables: Experiment registry, Backtesting framework, Replay engine, Research database, Strategy comparison tools, Performance analytics.

Success Criteria: Strategies can be tested and compared using historical data. Experiments are reproducible and versioned.

Phase 6 — Paper Trading

Objective: Validate the complete platform without financial risk.

Deliverables: Simulated trading, Virtual portfolio, AI performance tracking, Execution simulation, Risk validation, Performance reporting.

Success Criteria: Stable paper trading over an extended evaluation period. Consistent alignment between decisions and execution.

Phase 7 — Live Trading (Controlled Release)

Objective: Introduce real-market execution under strict safeguards.

Deliverables: Broker execution engine, Live order management, Position monitoring, Real-time alerts, Emergency stop mechanisms.

Success Criteria: Small-position live trading with strict risk controls. Reliable execution and complete audit logging.

Phase 8 — Continuous Learning

Objective: Enable systematic improvement of AEGIS.

Deliverables: Model performance evaluation, Strategy version comparison, Drift detection, Experiment recommendations, Knowledge Graph expansion, AI retraining workflows.

Success Criteria: Platform demonstrates measurable improvement over time through controlled updates.

Milestones

- Milestone 1 — Foundation complete.
- Milestone 2 — Reliable market intelligence operational.
- Milestone 3 — AI Council functioning.
- Milestone 4 — Risk Engine operational.
- Milestone 5 — Research Laboratory complete.
- Milestone 6 — Successful paper trading.
- Milestone 7 — First successful live trade.
- Milestone 8 — Continuous learning ecosystem operational.

Quality Gates

Before advancing to the next phase, each phase must satisfy: Unit testing completed, Integration testing completed, Performance testing completed, Documentation updated, Architecture reviewed, No unresolved critical defects.

Long-Term Expansion

Future versions may introduce: Multi-broker support, Stocks and ETFs, Cryptocurrency markets, Commodities, Reinforcement Learning, Portfolio optimization, Institutional order-flow analysis, Mobile companion application, AI-powered research assistant.

Project Success Metrics

The success of AEGIS will be measured by: System stability, Decision explainability, Risk management effectiveness, Research quality, Backtesting accuracy, Paper trading consistency, Live trading discipline, Continuous improvement over time.

Profitability is important but shall never outweigh engineering quality and capital preservation.

Closing Statement

The Development Roadmap provides the structured path from concept to production. By progressing through clearly defined phases with measurable success criteria, AEGIS can evolve into a robust, explainable, and adaptive Artificial Financial Intelligence Platform while minimizing technical debt and

reducing implementation risk.

MASTER PROJECT CONSTITUTION

Document ID: AEGIS-DOC-010

Version: 1.1

Status: Foundational Document

Classification: Constitutional Engineering Document

Project: AEGIS – Artificial Financial Intelligence Platform

Executive Summary

This document establishes the constitutional rules that govern Project AEGIS. It is the highest-level reference for the project's identity, architecture, engineering philosophy, and long-term direction.

Every future design decision, software implementation, AI model, database schema, and research activity shall align with this constitution unless formally revised.

1. Project Identity

Project Name: AEGIS

Meaning: Inspired by the legendary shield, AEGIS symbolizes protection, intelligence, resilience, and disciplined defense. In this platform, protecting capital is as important as identifying opportunities.

Project Type: Artificial Financial Intelligence Platform

Primary Domain: Forex (Version 1), with planned expansion to Stocks, Cryptocurrencies, Commodities, ETFs, Futures, and other financial markets.

2. Mission

To build an explainable, adaptive, and research-driven financial intelligence platform that assists trading decisions through scientific methodology, modular AI, disciplined risk management, and continuous learning.

3. Vision

To create an institutional-grade Artificial Financial Intelligence Platform capable of: Understanding market context, Coordinating specialized AI experts, Protecting capital through disciplined risk management, Continuously learning from experience, Producing transparent and explainable decisions.

4. Core Philosophy

AEGIS is not built to chase trades. AEGIS is built to understand markets.

The platform values: Knowledge before prediction, Evidence before opinion, Risk management before profit, Long-term consistency before short-term gain, Scientific research before assumptions.

5. Architectural Principles

The platform shall always remain: Modular, Explainable, Scalable, Secure, Observable, Fault tolerant, Testable, Version controlled.

Each subsystem shall have a single, clearly defined responsibility.

6. Intelligence Model

AEGIS uses a collaborative intelligence model. Core components include: AI Council, Chief Strategist, Research Director, Risk Engine, Execution Engine.

No single AI model is solely responsible for trading decisions. Decisions emerge through collaboration and validation.

7. Decision Pipeline

Every trading opportunity must follow this sequence: Observe → Validate → Generate Features → AI Council Analysis → Chief Strategist Recommendation → Risk Validation → Execution → Outcome Analysis → Knowledge Update → Continuous Learning.

No stage may be skipped.

8. Technology Foundation

Official Version 1 stack:

Frontend: React, TypeScript, Tailwind CSS.

Backend: Python, FastAPI.

Databases: PostgreSQL, TimescaleDB.

Infrastructure: Redis, NATS, Docker, DigitalOcean.

Machine Learning: PyTorch, XGBoost, LightGBM, Scikit-learn, MLflow.

Monitoring: Prometheus, Grafana, Loki.

9. Engineering Rules

Every contribution to AEGIS shall: Be documented, Be version controlled, Be testable, Be explainable, Be measurable, Be reproducible.

No undocumented production change shall be accepted.

10. Research Rules

Research shall follow the scientific method: Observation → Hypothesis → Experiment → Evaluation → Conclusion → Knowledge.

Every experiment must: Have a clear objective, Be reproducible, Preserve datasets, Record parameters, Document conclusions.

Failed experiments are considered valuable research.

11. AI Governance

Every AI model shall: Have a defined responsibility, Maintain version history, Track performance metrics, Produce explainable outputs, Preserve reasoning history.

Models shall be evaluated continuously and improved through controlled experimentation.

12. Risk Governance

The Risk Engine is the final authority for trade approval. No AI model, strategy, or administrator may bypass risk validation.

Capital preservation always has priority over opportunity.

13. Data Governance

All critical data shall be: Accurate, Auditable, Versioned, Secure, Backed up, Traceable.

Historical records shall remain immutable whenever practical.

14. Long-Term Objectives

AEGIS aims to become: A financial research platform, A quantitative analysis environment, A continuous learning ecosystem, A reusable AI architecture, A trusted decision support system for multiple financial markets.

15. Success Definition

The success of AEGIS is measured by: High-quality decision-making, Reliable risk management, Explainable AI, Stable architecture, Continuous learning, Research excellence, Sustainable long-term performance.

Profitability is important but shall never compromise engineering discipline or responsible risk management.

16. Founder Commitments

The Founder commits to: Prioritizing engineering quality over rushing features, Maintaining clear documentation, Making evidence-based decisions, Continuously learning and improving the platform, Preserving the architectural vision of AEGIS.

17. Architect Commitments

The Chief Systems Architect (Claude) shall: Protect architectural consistency, Recommend scalable and maintainable designs, Challenge assumptions when appropriate, Explain technical concepts clearly, Ensure new features align with the established vision, Promote engineering best practices throughout the project's evolution.

18. Constitutional Laws

The following principles shall not be violated without formal review:

- Risk Engine has final authority.
- Every decision must be explainable.
- Every AI model must have a defined responsibility.
- Every experiment must be reproducible.
- Every strategy must be versioned.
- Every production change must be documented.
- Every trade contributes to institutional knowledge.
- Architecture must remain modular.
- Security is mandatory.
- Continuous learning is a core capability.

Closing Declaration

Project AEGIS is founded on the belief that disciplined engineering, scientific research, explainable artificial intelligence, and responsible risk management can produce a financial intelligence platform that improves continuously over time.

This constitution serves as the guiding document for every future stage of development. As AEGIS evolves, new capabilities may be added, but they must reinforce—not replace—the principles established here. By adhering to this constitution, the project aspires to become a robust, transparent, and adaptive Artificial Financial Intelligence Platform capable of supporting informed financial decision-making across changing markets and technologies.